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Linux on the iBall Compbook

Join the author as he describes the challenges he faced when he used a Compbook to replace his netbook and then tried to make Linux work on it. He ruefully comments, "Making Linux work on a system with a Bay Trail Atom processor is still a work in progress."

My netbook failed and the cost of repairing it was expected to be greater than the newly released iBall Compbook. Besides, the Compbook weighs just one kilogram.

Someone suggested that Linux could be installed on it. Bay Trail Atom CPUs by Intel are widely used on Chromebooks and the ChromeOS uses Linux as its core. But still, I should have done more research before attempting to install Linux on the Compbook! The experience turned out to be unexpectedly complex and incomplete.

32-bit EFI BIOS

Fedora works fine with the EFI BIOS. I created the live USB and plugged it into the Compbook, with no luck at all! This led to the discovery that while the Bay Trail Atom CPU is 64-bit, it uses a 32-bit EFI BIOS. No distribution supports it out-of-the-box.

The simplest option is to install Grub2Win on Windows and make it the default firmware for booting. For this, the 'Secure Boot' option has to be disabled in the BIOS.

You will need to make space on the first disk for installing Linux. This can be done from Windows, before the Linux installation or during the installation.

CAUTION

Do not erase Windows from your system, as yet.

I found that I could not use the 32GB microSD card for installation as the BIOS does not recognise it. So, the inbuilt 32GB disk has to be used.

I used the Grub2 commands to boot from the live USB. The live USB created by 'Fedora Media Writer' could not be used as the BIOS could only find the EFI partition but not the partition containing the installation files. Hence, I used Unetbootin to create the live USB, setting the partition label to 'Live'. The following Grub commands were used for booting the live USB:

```
grub> ls
(hd0) (hd0,msdos1) (hd1,gpt5) ...
```

```
grub> ls (hd0,msdos1)/isolinux
boot.cat ... initrd0.img ... vmlinuz0 ...
grub> root=(hd0,msdos1)
grub> linux /isolinux/vmlinuz0 root=live:LABEL=Live ro rd.live.
image
grub> initrd /isolinux/initrd0.img
grub> boot
```

Kernel parameters can be picked up from *grub.conf* in the *efi/BOOT* partition. It takes a while for GNOME 3 to load.

The installation itself was fairly straightforward, except that it is probably better to create and specify the installation partition, manually. The partition label for the Linux root was set as 'Linux'.

'Rufus' worked well for creating a usable live USB on Windows.

Grub2Win configuration

For booting in linux, it is easier to add the Fedora boot option to the Grub2Win menu. The options are:

```
OS Type: fedora
Custom Script:
set root='(hd0,gpt5)'
linux /boot/vmlinuz-4.6.7-300.fc24.x86_64
root=LABEL=Linux verbose intel_idle.max_cstate=1
initrd /boot/initramfs-4.6.7-300.fc24.x86_64.img
```

The reason for the kernel option, 'intel_idle.max_cstate=1', is given in the section below.

If the kernel is updated, you will need to change the boot menu entry in Grub2Win.

Hard locks, system freezes and slow graphics

After Linux was installed, my happiness was short lived. The touchpad would stop working after a short time. However, a USB mouse worked well. But the more serious issue was that the system would freeze periodically for a considerable period of time.

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